Spread and box plots

Lesson questions

20 daily questions include C1 and C2. Use separate paper when helpful.

P1. Data are 3, 7, 8, 10, 12. Find the range.

P2. Find the five-number summary of 1, 3, 4, 6, 8, 9, 11. Exclude the median from each half.

P3. Find the five-number summary and IQR of 1, 2, 4, 6, 8, 10, 12, 13.

P4. Data are 2, 4, 6, 8. Find the mean, each distance from the mean, and the MAD.

P5. A box plot has five-number summary 0, 2.5, 5, 9, 12. Which interval represents the upper quarter? Does a longer interval mean more data?

P6. Compare A: 4, 4, 6, 6 and B: 1, 3, 7, 9. Find each mean and MAD. Which set is more spread out?

P7. Data are 1, 3, 4, 5, 6, 7, 9, 11. Replace 11 with 31. Compare the old and new range and IQR.

F. Data are 3, 5, 7, 9. Mean = 6. Distances are 3, __, __, 3. MAD = __.

D. Zoe averages −3, −1, 1, and 3 to find MAD. She gets 0. What should she average?

X1. Find the range of 2, 4, 6, 9, 12. Which values determine it?

X2. Find the five-number summary and IQR of 2, 3, 5, 7, 9, 11, 13. Exclude the median from each half.

X3. Find the five-number summary and IQR of 2, 4, 5, 7, 9, 11, 12, 14.

X4. Find the mean and MAD of 2, 4, 8, 10. Show all four distances.

X5. A box plot has summary (1, 3, 6, 10, 13). Find range and IQR, and name the interval containing the upper quarter.

X6. Compare MAD for A: 5, 5, 7, 7 and B: 2, 4, 8, 10. Both means are 6.

X7. Data are 2, 3, 5, 7, 9, 11, 13, 14. Replace 14 with 34. Compare the old and new range and IQR.

S. IQR measures __. MAD measures __.

E. For 1, 3, 3, 5, 7, 9, 11, 13, find Q1, the median, Q3, and the IQR. Show the two halves.

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Grade 6 cumulative check - assessment alternative

20 daily questions include C1 and C2. Use separate paper when helpful.

Use this in place of ordinary practice, or in a separate assessment sitting. It is not added to the 20 daily questions.

1. A picture shows 5 green tiles and 2 white tiles. Write green : white and white : green.

2. The pairs (2, 7), (4, 14), and (6, __) have the same ratio. Complete the table.

3. A bus covers 10 miles in 25 minutes at constant speed. How many miles in one hour?

4. Find 140% of 30 points.

5. 12 shells are 30% of a collection. Find the whole collection.

6. How many $\frac{3}{4}$-liter groups fit in $\frac{7}{8}$ liter? Include part of a group.

7. Share $\frac{5}{6}$ liter equally among 4 cups. How much goes in each cup?

8. Which is the reciprocal of $2\frac{1}{2}$: $\frac{2}{5}$ or $\frac{5}{2}$? Explain using a product of 1.

9. Three notebooks cost $2.45 each. What is the total cost?

10. Find $9.6 \div 8$.

11. Find $1{,}344 \div 32$. Verify by multiplication.

12. Expand $2(x + 7)$.

13. Solve $x + 9 = 23$.

14. Solve $(\frac{3}{4})\,x = 12$.

15. A parallelogram has base 9, height 4, and slanted side 5 cm. Find its area.

16. A triangle has base 9 cm and height 4 cm. Find its area.

17. A prism has right-triangle ends with sides 3, 4, 5 cm and length 4 cm. Find surface area.

18. Order -7, 2, -1 from least to greatest.

19. Reflect (-3, 4) across the y-axis.

20. Write an inequality for x that means at least 9. Is the boundary included?

21. Find the mean of 2, 3, 7, 8.

22. Find the median of 9, 1, 5, 3, 7.

23. Find Q1, Q3, and IQR for 1, 2, 4, 5, 7, 8, 10, 11. Exclude the median if the count is odd.

24. Find the MAD of 1, 3, 5, 7.

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Challenge and reasoning

20 daily questions include C1 and C2. Use separate paper when helpful.

C1. Compare A: 2, 4, 6, 8 and B: 0, 4, 6, 10. Find each mean, range, and MAD. Which has the same center but greater spread? Describe what changed in the data.

C2. For 2, 4, 6, 8, 10, 12, 14, find the five-number summary and IQR; exclude the median from each half. Add 3 to every value. Predict and verify the new summary, IQR, and range. Explain which measurements stay unchanged.

Worked example - compare with the slide model

Find the IQR for 2, 4, 6, 8, 10, 12.

Split the ordered data into 2, 4, 6 and 8, 10, 12.

The two half-medians are Q1 = 4 and Q3 = 10.

IQR = 10 − 4 = 6.

Result: 6